MATH210B Homework 6

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1 Problem 8

Let $A : V \to V$ be a linear operator. Let β be a basis such that $[A]_\beta$ is diagonal.

Then, $A(v) = \lambda v$ for any $v \in \beta$.

If this is the case, then, V is the direct sum of eigenspaces since eigenspaces with different eigenvalues don't intersect and since the eigenbasis generates V, they cover all of V.

Then, all elementary divisors of A are linear since the characteristic polynomial splits.

This immediately implies that all invariant factors of A are products of distinct linear polynomials since they are products of elementary divisors. Since the minimal polynomial is an invariant factor, this also immediately implies that the minimal polynomial is a product of linear polynomials.

Lastly, notice that cyclic modules $F[x]/(x - \lambda_i)$ correspond to subspaces where A acts by multiplication by λ_i, so if we can decompose V into such subspaces, then A is diagonalizable.

2 Problem 9

If $A^N = 0$ for some $N > 0$, x^N acts as zero. Then, the minimal polynomial divides x^N and is thus x^k for some $k \leq n$, which produces the desired result.

3 Problem 10

Notice that $det(A - tI) = det((A - tI)^t) = det(A^t - tI)$, so both matrices have the same characteristic polynomial. But then they have the same elementary divisors and they're thus similar.